Classical Field Theory — Exercise Sheet 17

Forced and accidental massless directions

Questions only

Use the real scalar kinetic coefficient $1/2$ and the orbit–Hessian lemma of Lecture 17. Counts of massless modes below refer to the linearised field. A vacuum background need not belong to L^2; perturbation data may be Schwartz or in the earlier energy class.

Exercise 17.1 (The full orthogonal orbit). For $N \geq 3$, $\lambda, v > 0$, consider

$$V(u) = \frac{\lambda}{8}(|u|^2 - v^2)^2, \qquad u \in \mathbb{R}^N, \qquad u_0 = vn, \quad |n| = 1,$$

with the usual $O(N)$ action and canonical kinetic term. Find the vacuum set, the stabiliser at u_0, its dimension, and the orbit tangent. Derive the Hessian and diagonalise the linearised field using the radial and tangential projectors, without first rotating n to a coordinate axis. Identify which masses are symmetry-forced. Finally discuss $N = 1$ separately: does its broken discrete symmetry force a massless mode?

Exercise 17.2 (A flat singlet still has a nonlinear restoring force). Add a real scalar z with its own canonical kinetic term and potential

$$\widetilde{V}(u, z) = \frac{\lambda}{8}(|u|^2 - v^2)^2 + z^4.$$

Let $O(N)$ act on u and trivially on z. Find the vacuum set, the Hessian kernel at $(vn, 0)$, and the subspace forced by the symmetry. Can the extra singlet direction be tangent to a curve of vacua? On the spatial torus, set $u = vn$ and take a homogeneous singlet $z(0) = A > 0$, $z_t(0) = 0$. Derive its nonlinear period as a definite integral and its dependence on A. Explain how a zero linearised mass is compatible with this nonlinear oscillation. These homogeneous data are not claimed to be L^2 data on $\mathbb{R}^d$.

Exercise 17.3 (An ineffective group factor belongs to the stabiliser). Take $G = SO(3) \times U(1)$ acting on a real triplet by $(R, e^{i\theta})u = Ru$. Use the radial potential of Problem 1 with $N = 3$ and vacuum $u_0 = v(2, -1, 2)/3$. Determine the full stabiliser and the differential of the orbit map $(c, \theta) \mapsto c \times u_0$ on $\mathbb{R}^3 \oplus \mathbb{R}$. Find a basis of its kernel, the number of broken directions and the Hessian nullity. Why would counting $\dim G$ or only counting rotations that fix u_0 give a wrong answer?

Exercise 17.4 (Kinetic normalisation before diagonalising masses). For a real three-component perturbation h, consider the quadratic density

$$L = \frac{1}{2}\partial_\mu h^{\mathsf{T}} K \partial^\mu h - \frac{1}{2} h^{\mathsf{T}} M h, \qquad K = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 9 \\ 0 & 9 & 9 \end{pmatrix}.$$

The real phase data may be taken Schwartz. Verify the factorisation $K = S^{\mathsf{T}} S$ for

$$S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Derive the canonical momentum, and find the corresponding change of coordinates and momenta that preserves $\int \delta h_a \wedge \delta p_a$. Compute the physical squared masses and compare them with the raw eigenvalues of M. State the general eigenproblem and explain what happens to the kernel dimension under this kinetic normalisation.